

**B. Tech. (CSE)****Course: Mathematics-I****Date of CA3: 13/11/2025****Semester I (Sections A, B, C, D, E & F)****CA-3: Assignment Test (Partial Derivatives)****Max. Marks: 10****Time: 45 Min.**

1. If $u = \sin^{-1}(x/y) + \tan^{-1}(y/x)$, find the value of $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y}$.
2. If $u = \frac{x}{y+z} + \frac{y}{z+x} + \frac{z}{x+y}$, show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = 0$.
3. If $u = \frac{1}{\sqrt{x^2+y^2+z^2}}$, prove that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$.
4. If $u = x^y$, show that $\frac{\partial^3 u}{\partial x^2 \partial y} = \frac{\partial^3 u}{\partial x \partial y \partial x}$.
5. If $u = \log(x^3 + y^3 - x^2y - xy^2)$, then show that $\frac{\partial^2 u}{\partial x^2} + 2 \frac{\partial^2 u}{\partial x \partial y} + \frac{\partial^2 u}{\partial y^2} = -4(x+y)^{-2}$.
6. If $z = x\phi\left(\frac{y}{x}\right) + \psi\left(\frac{y}{x}\right)$, prove that $x^2 \frac{\partial^2 z}{\partial x^2} + 2xy \frac{\partial^2 z}{\partial x \partial y} + y^2 \frac{\partial^2 z}{\partial y^2} = 0$.
7. If $u = x^2 \tan^{-1}\left(\frac{y}{x}\right) - y^2 \tan^{-1}\left(\frac{x}{y}\right)$, evaluate $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2}$.
8. If $u = \log(x^4 + x^2y^2 + y^4)$, then show that $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} + 4 = 0$.
9. If $u = \tan^{-1}\left(\frac{y^2}{x}\right)$, then find the value of $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2}$.
10. If $u = \tan^{-1}\left(\frac{x^3+y^3}{x-y}\right)$, then prove that (i) $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \sin 2u$
and (ii) $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = 2\cos 3u \cdot \sin u$.
11. If $x = uv$ and $y = \frac{u+v}{u-v}$, find $\frac{\partial(u,v)}{\partial(x,y)}$.
12. If $u = \frac{yz}{x}$, $v = \frac{zx}{y}$, and $w = \frac{xy}{z}$, find $\frac{\partial(u,v,w)}{\partial(x,y,z)}$.



13. If $u = x + y + z$, $uv = y + z$, and $uvw = z$, find $\frac{\partial(u,v,w)}{\partial(x,y,z)}$.

14. If $u = \frac{x+y}{1-xy}$ and $v = \tan^{-1} x + \tan^{-1} y$, find $\frac{\partial(u,v)}{\partial(x,y)}$.

Are u and v functionally related? If so, find the relation between them.

15. If $u = xy + yz + zx$, $v = x + y + z$, and $w = x^2 + y^2 + z^2$, find $\frac{\partial(u,v,w)}{\partial(x,y,z)}$.

Are u, v , and w functionally related? If so, find the relationship between them.

16. If $u = y + z$, $v = x + 2z^2$, and $w = x - 4yz - 2y^2$, find $\frac{\partial(u,v,w)}{\partial(x,y,z)}$.

Are u, v , and w functionally related? If so, find the relationship between them.